What drives bottled-water sales, and what is media really worth?

Marketing mix model · 161 weeks of data, Jan 2022 – Jan 2025

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Weather, price and distribution drive sales; media adds about 1% and returns over £1 per £1 in only two channels

20%

of volume comes from warm weather: +1.9% per °C, faster above 20°C

+3.7%

volume for each +1 point of distribution, the strongest lever (about £6M a year)

−4.3%

volume when our price rises 10% vs competitors; our premium grew from +46% to +89%

~1%

of volume driven by media; £1.53 revenue per £1 across modelled channels

£3.07

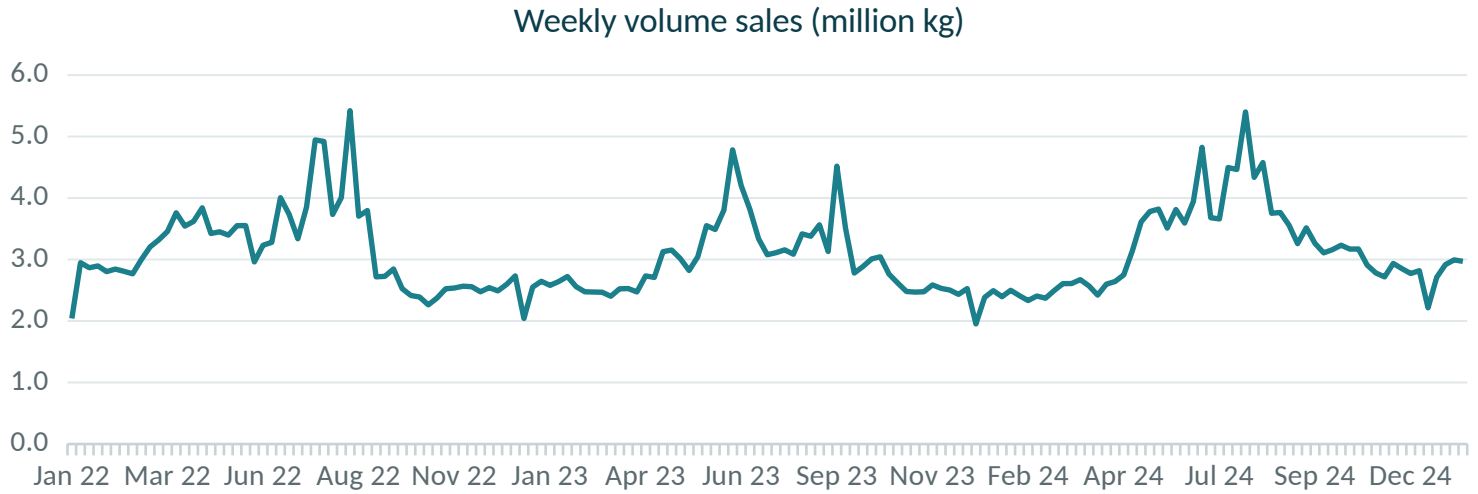
per £1 from Online partnership, the best proven channel; TV and online video under £0.30

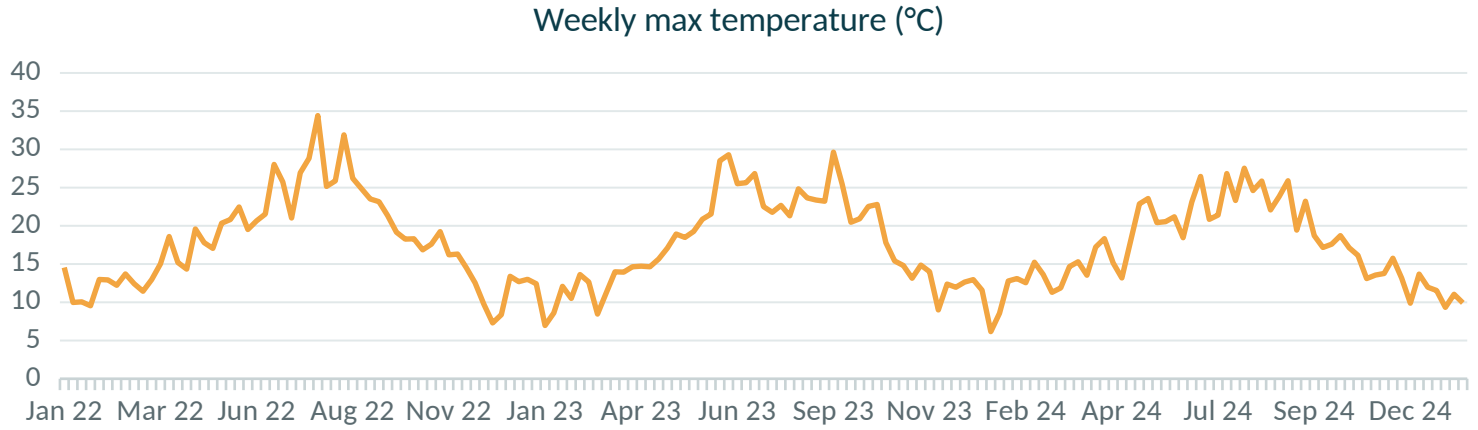
28%

of promoted volume is estimated to be extra; discounts deepened from 22% to 46% with no extra volume

Next moves: grow distribution · mind the price gap · shift TV/video budget to Online partnership · test OOH · shallower promotions

Three years of weekly data: sales follow the temperature, prices nearly doubled and media was sparse





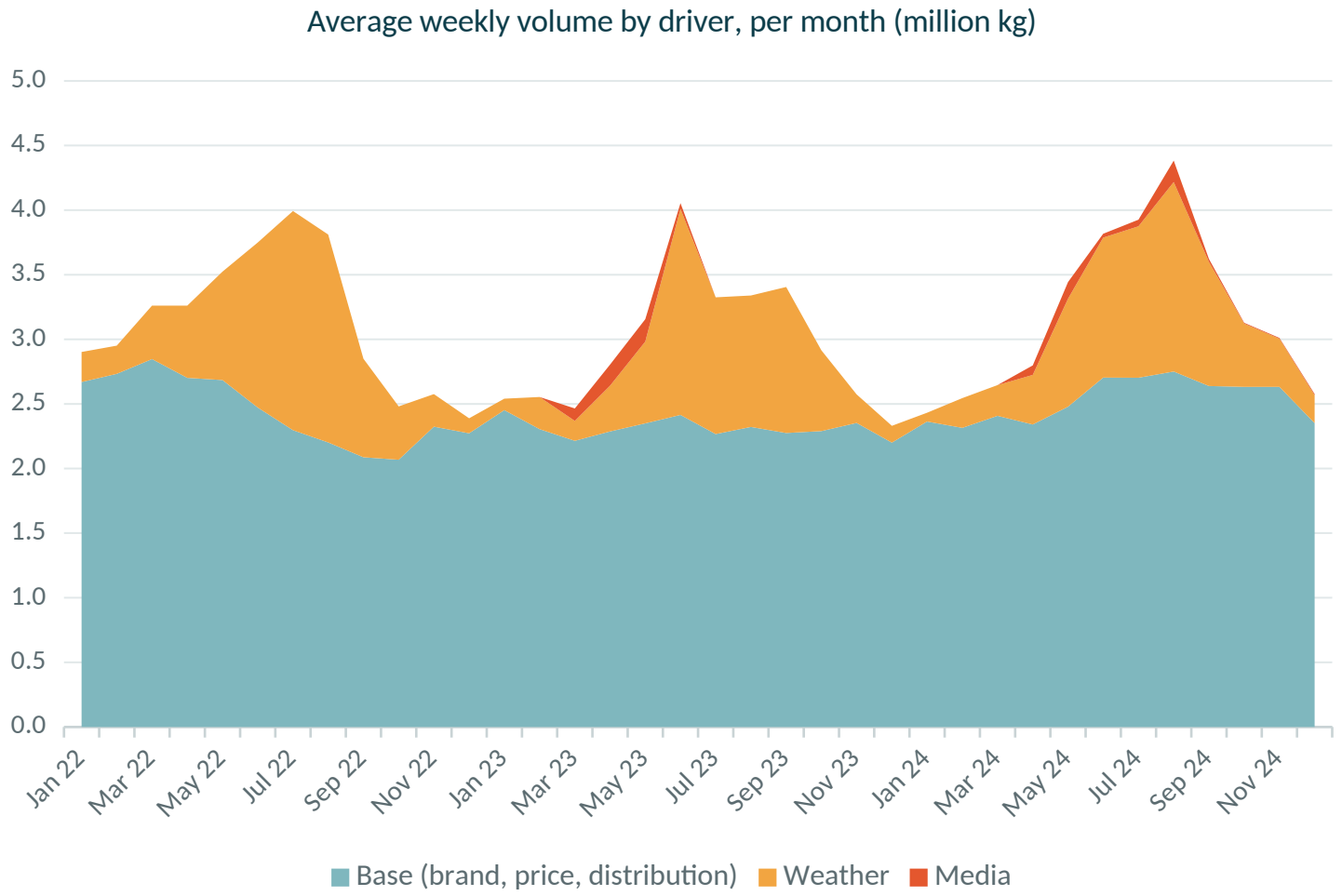
161 weeks
90 variables: sales, price, promotion, distribution, competitors, weather, media

£450M
value sales; volume moves with temperature (correlation 0.79)

£0.77 → £1.08
average base price per kg, 2022 to 2024

£3.5M
media over 3 years, only 0.8% of sales; TV ran 3 weeks, OOH 5

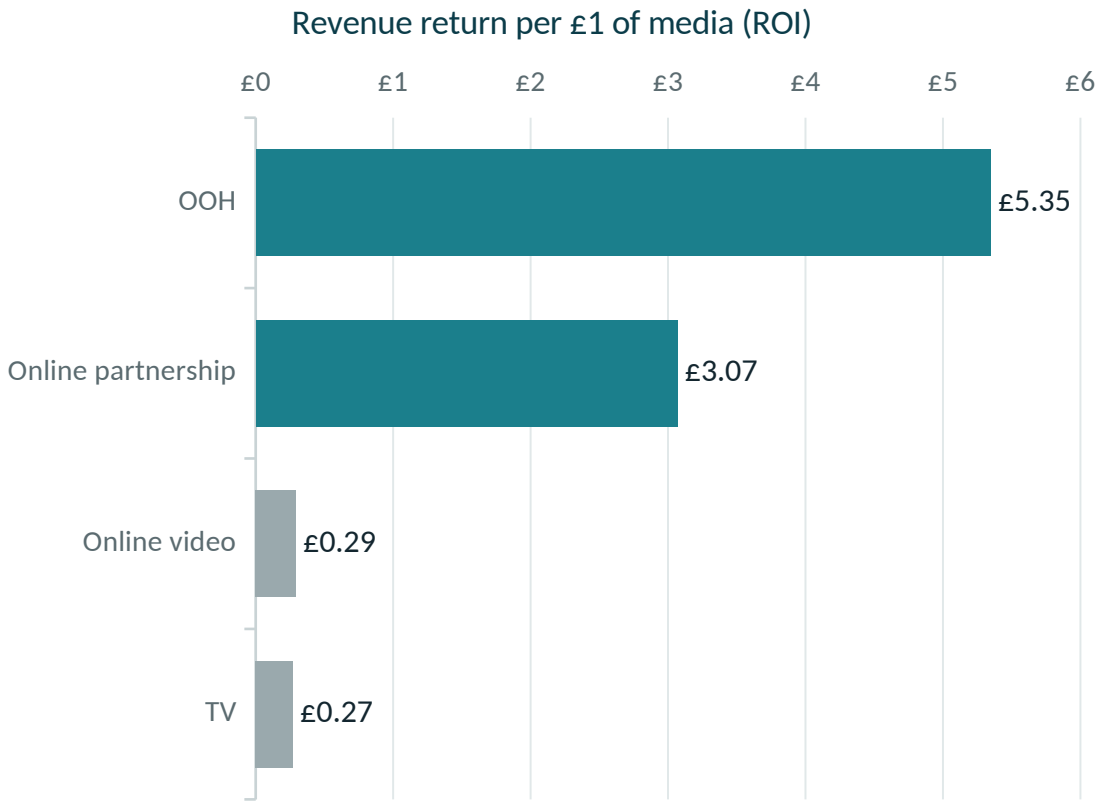
Base sales and weather explain 99% of volume; media adds a thin layer of about 1%



Driver	Effect
+1°C max temperature	+1.9%
Each °C above 20°C (extra)	+2.2%
+10 mm rain	−1.7%
Christmas / New Year week	−22%
+10% price vs competitors	−4.3%
+1 point distribution (ACV)	+3.7%

All effects highly significant (|t| > 4). Base = sales in a cool, dry week without media, at that week's price and distribution.

Online partnership returns £3.07 per £1; TV and online video returned under £0.30

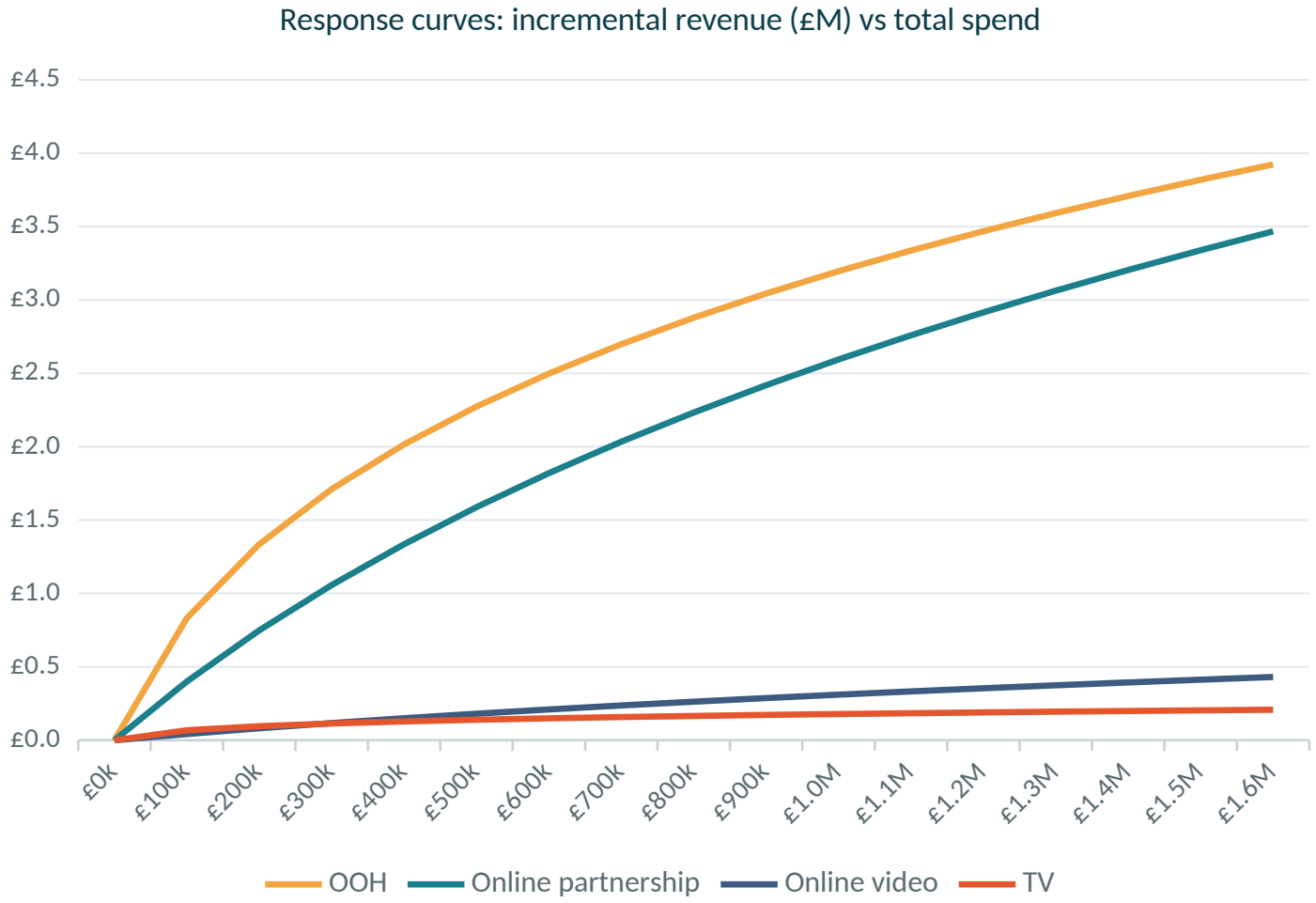


Channel	Spend	ROI	95% range	Evidence
OOH	£350k	**£5.35**	£0.00–£13.18	Medium
Online partnership	£575k	**£3.07**	£1.65–£4.52	High
Online video	£1,250k	**£0.29**	£0.00–£5.42	Low
TV	£529k	**£0.27**	£0.00–£0.90	Low

Read the range, not just the bar. OOH looks best but rests on 5 weeks of activity, so its range starts at zero. Online partnership is the only channel that is both high and certain.

Not measurable: Social (£736k) showed no positive lift; Search (£19k) and RDM (£10k) are too small to estimate. ROI = incremental revenue ÷ spend (revenue, not profit: no margin data).

Moving 30% of TV and video budget into Online partnership could add about £0.9M at the same spend



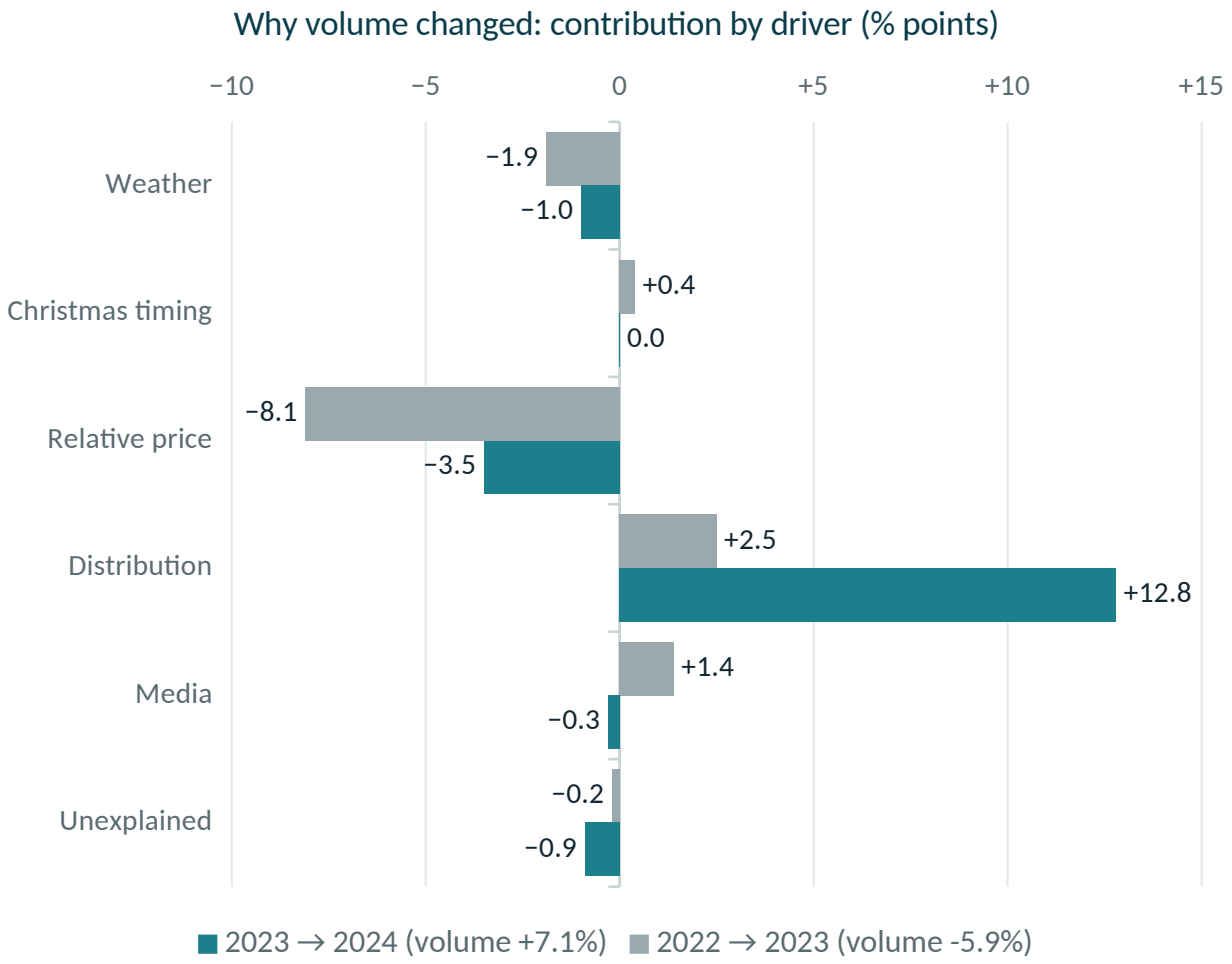
Channel	Next £1 returns	Spend now
OOH	£2.95	£350k
Online partnership	£2.22	£575k
Online video	£0.20	£1,250k
TV	£0.10	£529k

+£0.9M revenue

Shift £534k from TV and online video into Online partnership, same total budget

TV spent £176k a week for 3 weeks, well past saturation: if kept, spread it over more weeks. Test OOH before scaling it.

A wider price gap cost volume in 2023; distribution gains won it back in 2024; deep promotions mostly subsidise sales



Price

Our base price premium over competitors grew from +46% (2022) to +89% (2024). Elasticity −0.46: the 2023 rise cost about 8% volume.

Distribution

ACV rose from 39.9 to 43.5 in 2024, worth about +13% volume. We still reach under half of the market.

Promotions

Promoted share of volume went from 15% to 34%. Promotions lift sales by widening store reach, not through discount depth; an estimated ~28% of promoted volume is extra.

A simple, transparent regression that forecasts unseen weeks within 4%, with clear limits

1 Clean and explore

Dates, gaps, definitions; 18 charts; traps found (e.g. store counts follow demand)

2 Transform media

Carryover (adstock) and diminishing returns (saturation) per channel

3 Regress

Log of weekly volume on 10 drivers, robust (Newey-West) errors

4 Validate

12-week holdout, residuals, multicollinearity, leave-one-year-out

0.91 adjusted R^2: share of weekly variation explained

4.5% average weekly error

4.2% error on 12 unseen weeks

< 4 variance inflation for every driver: no harmful overlap

How far to trust it

- TV ran 3 weeks and OOH 5: their ROIs have wide ranges
- Channels overlap in time (TV inside the Online partnership burst)
- Revenue ROI, short-term only: no margins, no long-term brand effect
- Price effect shrinks if a time trend is added (−0.46 vs −0.10)
- Distribution counts stores that sold, so it can partly follow demand

Recommendations: grow reach, mind the price gap, rebalance media, rethink promotions

1

Protect and grow distribution

Each +1 point of ACV is worth about £6M a year; we reach under half the market. Target the missing retailers.

2

Manage the price gap

Our premium over competitors reached +89%. Price against competitors, not in isolation; test before further rises.

3

Rebalance media

Move TV and video money to Online partnership; run a regional test of OOH; if TV stays, spread it over more weeks.

4

Rethink promotions

Test shallower discounts (25–30%) and focus on store reach rather than depth.

Questions for the client

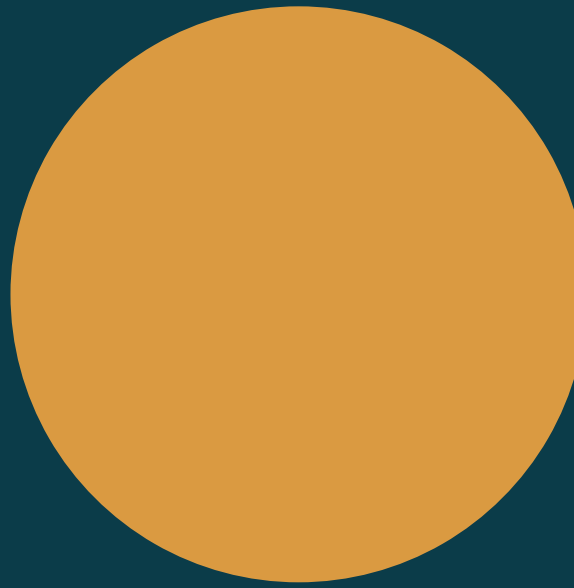
- Price per kg nearly doubled: list price rises or a shift to smaller packs?
- Why were promotions paused from June to December 2022?
- What exactly are Online partnership and RDM?
- Is other marketing (in-store, trade) missing from the data?

Next steps for the model

- Bayesian MMM with priors
- Regional data and lift tests
- Margin data for profit ROI

Appendix

Model details, diagnostics, media calendar and variables tested



Model specification: log-linear OLS on weekly volume, 10 drivers

ln(volume) = b0 + weather + Christmas + b·ln(price / competitor price) + b·ACV + Σ b·ln(1 + adstock(spend) / scale)

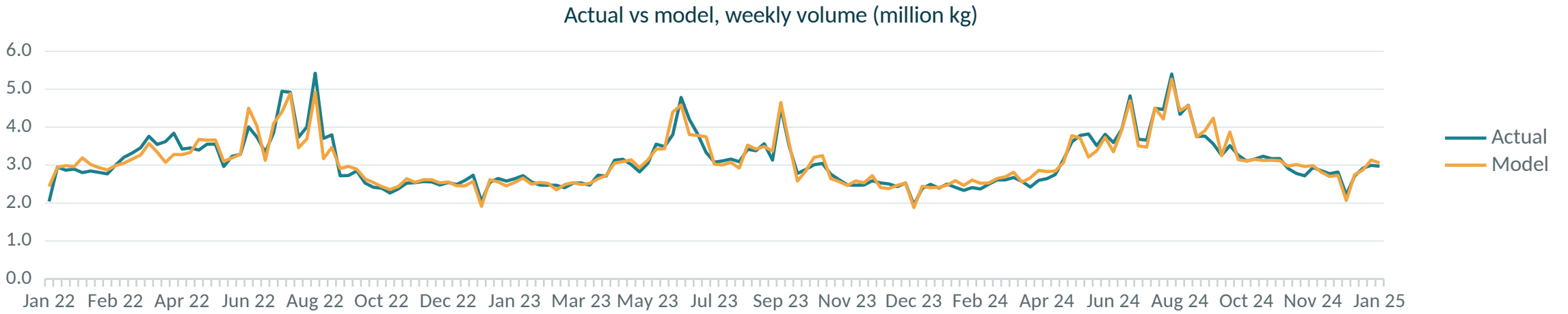
Driver	Coefficient	t-stat (HAC)	Evidence
Max temperature (°C)	0.0187	11.9	High
Degrees above 20°C	0.0221	5.1	High
Rainfall	-0.0017	-5.8	High
Christmas / New Year week	-0.2426	-4.8	High
ln(base price / competitor price)	-0.4569	-6.0	High
Weighted distribution (ACV %)	0.0364	9.8	High
TV	0.0037	0.9	Low
Online video	0.0037	0.1	Low
Online partnership	0.0754	4.3	High
OOH	0.0267	1.4	Medium

Weekly carryover rates

- TV: 0.3
- Online video: 0.8
- Online partnership: 0.0
- OOH: 0.5

Grid search within business ranges; best AIC among models with all media effects positive. Saturation scale = average adstocked spend in on-air weeks.

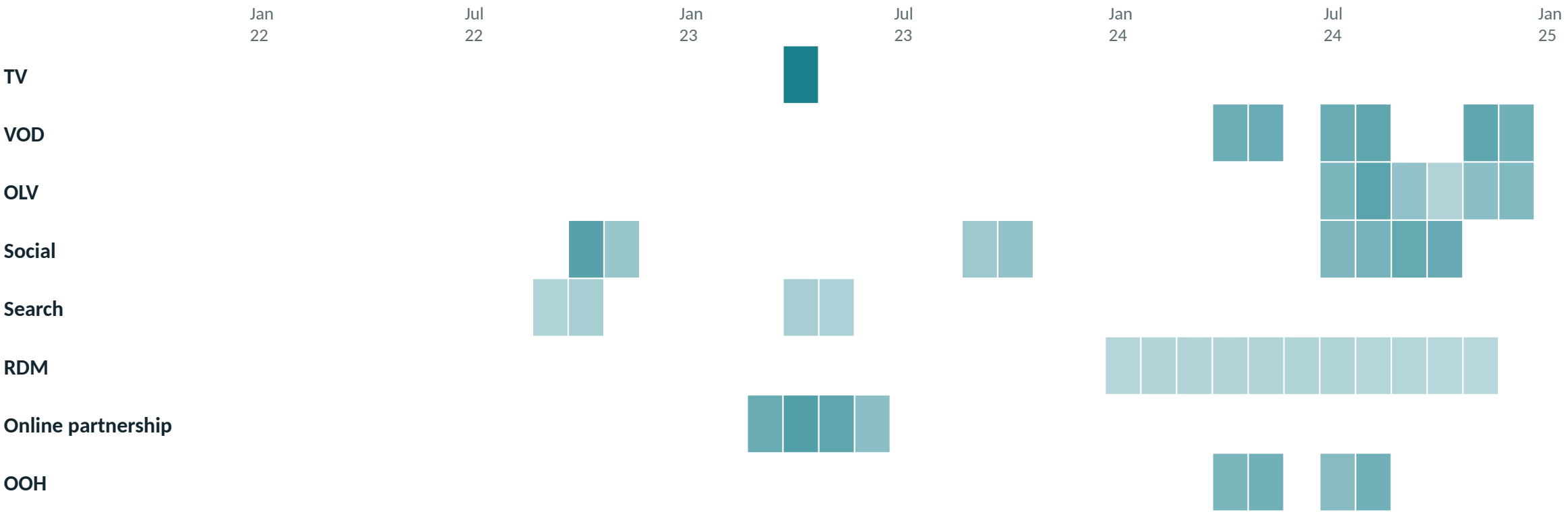
Model building and diagnostics: each driver group earns its place; residuals checked



Stage	Adj. R^2	Error
1. Temperature only	0.607	10.3%
2. + hot weeks, rain, Christmas	0.722	8.5%
3. + relative price	0.721	8.5%
4. + distribution	0.908	4.7%
5. + media (final model)	0.913	4.5%

Check	Value	p-value
Durbin-Watson	1.26	–
Breusch-Godfrey (4 lags)	21.80	0.000
Breusch-Pagan	24.88	0.006
Jarque-Bera	7.87	0.020
Residual std (log points)	0.06	–

Media calendar: sparse, bursty activity with channels overlapping in time



Darker = more spend that month. TV ran inside the Online partnership burst (spring 2023); in summer 2024 VOD, OLV, Social and RDM all ran together, during hot weather and heavy promotion. That overlap is why VOD and OLV are combined as Online video.

Variables tested and left out, and why

Variable	Coef.	t	Why it is out
Promo ACV distribution	0.0000	0.0	No effect once distribution is in; promotions work through store reach
Promo discount depth (%)	-0.0004	-0.8	No effect: deeper discounts did not add volume
Linear trend (per year)	-0.0627	-2.6	Collinear with relative price and has no business meaning (sensitivity only)
Competitor A media	0.0177	2.6	Positive (category halo?) but overlaps our 2024 video burst
Competitor B media	0.0067	0.7	No effect
Social media	-0.0089	-1.0	Negative sign in every specification: not measurable
Search media	-0.0191	-2.5	1% of budget; spurious negative sign
RDM media	-0.0029	-0.1	Under 1% of budget; no effect